Tianer Zhou

tianer.zhou@epfl.ch · (+41) 77 283 2714 · in/tianer-zhou · ezhoureal · ezhoureal.github.io

Education

EPFL (École Polytechnique Fédérale de Lausanne), Switzerland — Sep 2026 – Expected Summer 2028
M.Sc. in Computer Science **Coursework:** Algorithms II, Multiprocessor Architecture, Concurrent Computing

University of Michigan, Ann Arbor, USA — 2019 – 2022
B.S. in Computer Science, Summa Cum Laude **GPA:** 3.92/4.0
Coursework: Machine Learning, Operating Systems, Distributed Systems, Computer Organization, Database, Linear Algebra

Experience

Parallel System Architecture (PARSA) Lab, EPFL, Lausanne — Sep 2026 – Present
Student Researcher Server resource allocation for agentic AI workloads.

- **Workload Characterization:** Profiling agentic benchmarks (SWE-bench, DABstep, MLE-bench) served by Kimi K3 on the CSCS cluster at system and micro-architecture level, decomposing end-to-end task time into LLM inference, tool execution and harness overhead.
- **Resource Balance Studies:** Swept agent concurrency, CPU core count / frequency and cache provisioning; found today's servers oversupply CPU for agentic workloads, with throughput saturating at 2 cores per GPU.

Zhuhai Mingyue Private Fund Management, Remote — Apr 2026 – Present
Quantitative Developer (Part-time) Chinese equity & futures manager, 10B+ CNY AUM.

- **Event-Driven Trading System:** Built core components of a Python event-driven engine (signal monitoring, rebalancing, order management) bridging live market data and broker execution over Redis Pub/Sub for ∼5,000 stocks on 3 s snapshots; 10 signals deployed in pilot.
- **Reliability & Security:** Made the event bus thread-safe, replaced pickle with a versioned JSON codec, sandboxed user-submitted code, fixed look-ahead bias in the backtester; led a codebase audit (30 issues triaged P0–P3) and set up uv / ruff / pyright CI.
- **Data Service:** Hardened the firm's market-data API (FastAPI over ClickHouse / MySQL, ∼3B rows of minute bars) with LRU cache invalidation, negative-TTL caching, client timeouts and retries, and reproducible Docker builds.

Huawei Technologies, Hangzhou — Aug 2022 – Aug 2026
Graphics Engineer HarmonyOS graphics foundation, 80M+ devices.

- **On-Device Diffusion Model:** Quantized and deployed a 6B-parameter diffusion model on flagship mobile NPUs, generating 2K-resolution images in 10 s; fine-tuned it with LoRA on 4 V100s using FSDP on a curated 100+ image style-transfer dataset.

UI Framework Engineer C++ engine of the HarmonyOS native UI framework (ArkUI), 20,000+ apps.

- **#1 Contributor:** 1,200+ commits to the open-source C++/TypeScript UI engine, ranked first of 1,300+ contributors. [repo]
- **Layout Performance:** Optimized virtual scrollers with lazy loading, idle-time prefetching, list-item recycling and measure-result caching, cutting scroll jank by 60%.
- **Memory & Rendering:** Designed a deduplicated image cache saving 1 MB+ in typical icon-heavy workloads (10 MB+ at the extreme); built a React Native SVG renderer 20% faster than the Android implementation.

AI Systems Projects

- **GPT Training / Inference Stack:** Built GPT-style training and inference from scratch: BPE tokenizer, Transformer blocks, RoPE, FlashAttention and a distributed KV cache. [repo]
- **CUDA Kernels:** FlashAttention forward / backward in CUDA (SRAM tiling, online softmax, warp shuffles) and Triton; Tensor Core (WMMA) and int8 GEMMs benchmarked against cuBLAS / CUTLASS. [repo]
- **Open-Source LLM Inference (NVIDIA Dynamo):** Merged a fix for a potential out-of-bounds access in the Rust KV-router scheduler by deriving worker IDs from endpoints instead of a redundant ID vector. [PR #420]
- **World-Model Planning:** Optimized the planning loop of a hierarchical JEPA world model for up to 10x speedup; trained a subgoal planner that raised long-horizon success on PushT from 5% to 72%. [repo]
- **LLM Agent Tooling:** Built an MCP server that lets LLM agents inspect and control HarmonyOS device UI. [repo]

Technical Skills

Languages: C++17, Python, TypeScript, Rust · **AI / ML:** PyTorch, CUDA, Triton, FSDP, LoRA, quantization, MCP · **Systems:** multithreading, atomics, caching, tracing & profiling (ftrace, SmartPerf) · **Backend & Data:** FastAPI, Redis, ClickHouse, MySQL, Docker, CI (uv / ruff / pyright)